Section 4.7 Bijections & structural counting

双射与结构计数

[AIME-2026-II-Q02] | Part A (Q1-Q5)

A row of ten unit squares has each rising diagonal drawn. A bug travels along edges and diagonals from the bottom-left endpoint to the top-right endpoint, never reusing a segment or moving left along a nonvertical segment. If there are N paths, find $\sqrt{N}$.

十个单位正方形排成一行，各画从左下到右上的对角线。虫子沿边或对角线从整行左下角走到右上角，不重复线段，且非竖直步不得向左。若路径数为 N，求 $\sqrt{N}$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 4.7 Bijections & structural counting

双射与结构计数

[AIME-2005-II-Q06] | Part B (Q6-Q10)

Cards numbered 1 through $2n$ are stacked top to bottom, split into top half A and bottom half B, then alternately removed from B, A and added on top of a new stack. A stack is magical when at least one card from each half retains its original top-based position. Find $2n$ for the magical stack in which card 131 stays fixed.

编号 1 至 $2n$ 的牌自上而下排列，分为上半堆 A、下半堆 B，再依次从 B, A 顶端交替取牌并压到新堆顶。若原两堆各至少一张牌位置不变，称此堆神奇。求第 131 张保位的神奇牌堆总张数。

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Section 4.7 Bijections & structural counting

双射与结构计数

[AIME-2006-II-Q08] | Part B (Q6-Q10)

An equilateral triangle is subdivided by its three midsegments into four small congruent triangles. Color each small triangle uniformly with one of six colors. Up to rotations and reflections of the large triangle, count distinct colorings.

等边三角形用三条中位线分成四个全等小三角形，每个用六种颜色之一染成单色。将整体旋转、翻转视为相同，求不同染色数。

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Section 4.7 Bijections & structural counting

双射与结构计数

[AIME-2010-I-Q10] | Part B (Q6-Q10)

Count the quadruples of integers $0 \leq a_0, a_1, a_2, a_3 \leq 99$ satisfying $a_0 + 10a_1 + 100a_2 + 1000a_3 = 2010$.

求满足 $0 \leq a_0, a_1, a_2, a_3 \leq 99$ 且 $a_0 + 10a_1 + 100a_2 + 1000a_3 = 2010$ 的整数四元组数。

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Section 4.7 Bijections & structural counting

双射与结构计数

[AIME-2010-II-Q10] | Part B (Q6-Q10)

How many integer-coefficient quadratic polynomials have two integer roots, counting multiplicity, and constant term 2010?

求常数项为 2010、两个根均为整数（计重数）的整系数二次多项式个数。

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Section 4.7 Bijections & structural counting

双射与结构计数

[AIME-2011-II-Q06] | Part B (Q6-Q10)

How many integer quadruples $1 \leq a < b < c < d \leq 10$ satisfy $a + d > b + c$?

求满足 $1 \leq a < b < c < d \leq 10$ 且 $a + d > b + c$ 的整数四元组数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 4.7 Bijections & structural counting

双射与结构计数

[AIME-2018-I-Q10] | Part B (Q6-Q10)

Two concentric pentagonal rings have corresponding vertices joined by spokes. From an inner vertex a bug may step one vertex counterclockwise on that ring or outward on its spoke; from an outer vertex it may step clockwise or inward. Starting at a fixed inner vertex A, count fifteen-step returns to A, modulo 1000.

两个同心五顶点环由对应辐条相连。虫在内环可逆时针沿环一步或沿辐条向外，在外环可顺时针沿环一步或向内。由固定内顶点 A 出发，求十五步返回 A 的路径数模 1000。

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Section 4.7 Bijections & structural counting

双射与结构计数

[AIME-2023-II-Q10] | Part B (Q6-Q10)

Fill a labeled 2×6 grid bijectively with $1, \ldots, 12$, requiring neighboring cells that share an edge to have differences not divisible by 3. If there are N fillings, find the number of positive divisors of N.

将 1 至 12 各用一次填入有固定位置的 2×6 网格，要求共边两格的数之差不能被 3 整除。设填法数为 N，求 N 的正约数个数。

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Section 4.7 Bijections & structural counting

双射与结构计数

[AIME-2024-II-Q09] | Part B (Q6-Q10)

Using up to 25 identical white chips and 25 identical black chips, fill some cells of a 5×5 grid, at most one per cell. Every occupied row and column must be monochromatic. Count the placements to which no further chip can be added legally.

用最多 25 枚相同白棋子、25 枚相同黑棋子填入 5×5 方格，每格至多一枚。同一行、同一列内已有棋子均须同色。求无法再合法添入任何一枚棋子的布局数。

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Section 4.7 Bijections & structural counting

双射与结构计数

[AIME-2026-II-Q13] | Part C (Q11-Q15)

Two finite integer sets are related if they are disjoint, have the same size, and admit a pairing in which paired integers differ by one. Find the smallest possible size of a set related to exactly 4040 other sets.

若两个有限整数集合不交、元素数相等，且可配成每对整数相差一，则称它们相关。求恰与 4040 个其他集合相关的集合，其最小元素数。

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